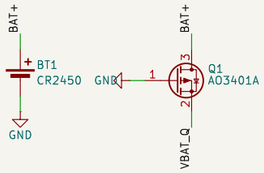
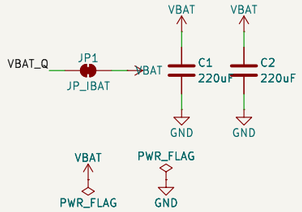
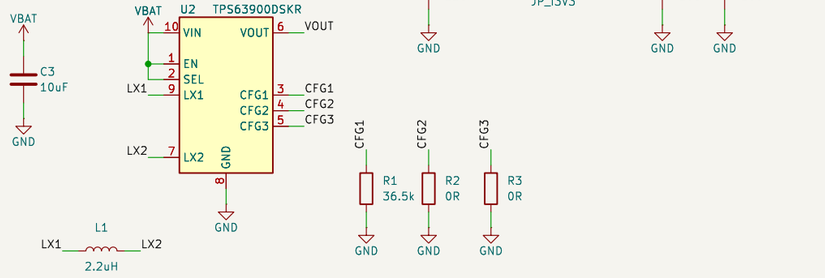
1 Battery CR2450, reverse polarity protection, bulk capacitors

Q1: drain to the cell, source to the load, gate to GND (a reversed cell turns it off)



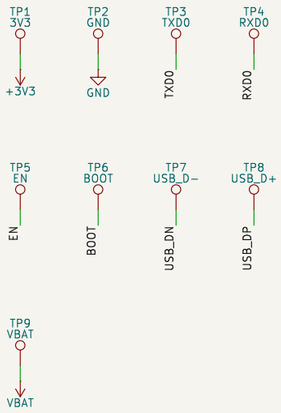


2 TPS63900 buck–boost, 3.3 V out, 75 nA quiescent



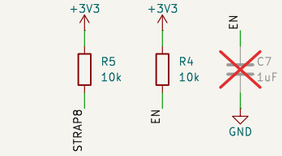
CFG1 36.5k + CFG2 0R: VO(2) = 3.3 V, no input current limit; CFG3 0R: VO(1) = 1.8 V, unused: EN, SEL to VIN

4 Test pads (bottom side, pogo pins)



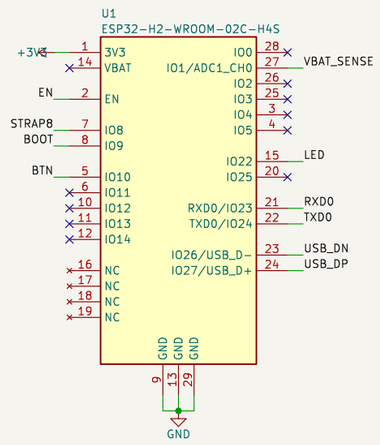
5 EN pull–up, strapping

C7: not fitted (the module has its own 10k/1uF RC)

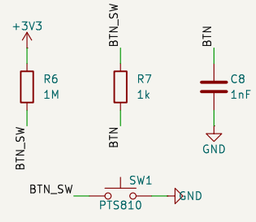


3 ESP32–H2–WROOM–02C

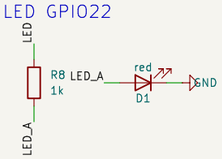
Pin 14 VBAT: leave open, tied to 3V3 inside the module



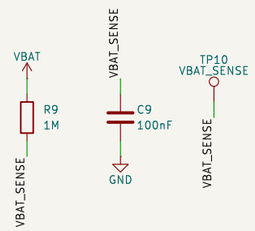
6 Button GPIO10 (LP IO, EXT1 wake–up)



7 LED GPIO22



8 Battery sense GPIO1 (ADC1_CH0), no divider



Generated by ../scripts/gen_sch.py from design.py
CR2450 -> Q1 (reverse polarity) -> TPS63900 3.3 V -> ESP32–H2–WROOM–02C
Haxeko

Sheet: /
File: matter–button–module–cr2450.kicad_sch

Title: ESP32–H2 Matter over Thread button, CR2450

Size: A3	Date: 2026–11–11	**Rev: 1.0**
KiCad E.D.A. 9.0.9		Id: 1/1